

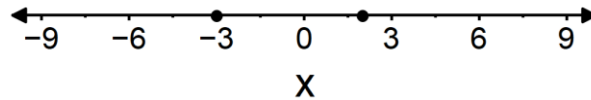
Graphical representations and origin of mathematical statements

Author: Cecilia Alvares

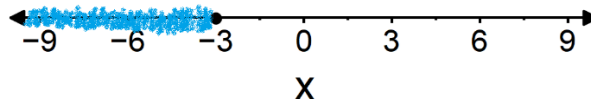
1) Graphical representation

The solutions of any mathematical statement (inequations and equations, functions included) can be represented in a more artistic form. This is achieved through what is called a **graphical representation**. There are several ways to do a graphical representation, but the most traditional and known one is by using a system of Rectangular coordinates. In this case, a set of three axis (maximum) are put together in a way to be orthogonal to one another, with each axis numerated according to a scale. If the statement contemplates a single unknown, the graphical representation would be made out of a graduated single axis. In this case, a point (which here is rather a small circle and not an element with no length in any direction) is indicated in the value(s) that corresponds to the solution(s). If the solutions are infinite as is the case of inequations, you may scribble the axis starting from a given value to fill the range of solutions concerned as an indication. The starting value is marked with a circle if it is included as a solution or with an open (i.e. not filled) circle if it is not included in the solution.

Practical example 1: given an equation containing one unknown which is found to have solutions $x=2$ and $x=-3$, we sketch:

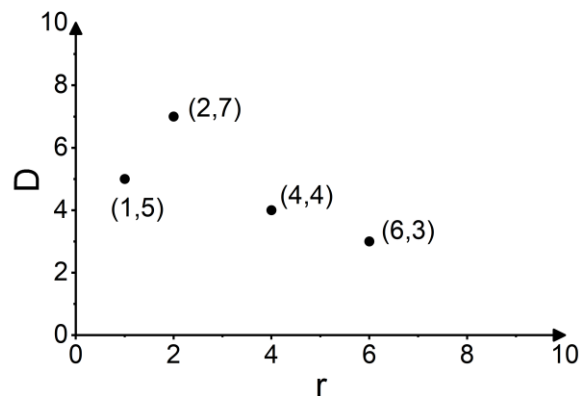


Practical example 2: given an inequation that has as a solution all $x \leq -3$, we sketch:



If the mathematical statement has two or three variables, two or three graduated axes are used, respectively, to make a graphical representation of the solutions. In the latter case, each axis is used to denote a variable. Then, for each combination that solves the statement, imaginary lines perpendicular to the axes associated to each variable at their value featured in the given combination are pictured and a point is drawn in the intersection of the perpendicular lines. This is done for each combination.

Practical example: imagine there is an equation relating two variables, D and r which has as solutions only the following combinations: $(r = 1, D = 5)$, $(r = 2, D = 7)$, $(r = 4, D = 4)$, $(r = 6, D = 3)$. By assigning the horizontal axis to r and the vertical axis to D , the graphical representation of these solutions would be:



Note that the indication of the variable values is optional.

It is important to stress the importance of having the number of variables in the statement not surpass 3 since otherwise it is not possible to do an actual **visual** representation in that way, although the idea of a graphical representation can be abstractly extrapolated to any statement.¹ Another limitation to note is that if the number of solutions

¹ In some cases where the statement has more than three variables, it is common to fix values for some of them so that only three remain, case in which it is possible to make an visual graphical representation for the “constrained statement”.

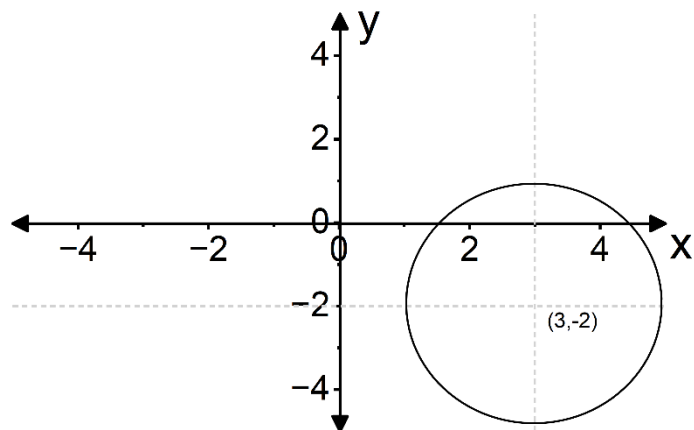
are infinite, it is evidently not possible to make a complete graphical representation (nor list them, for example) as infinite time would be required. In this case, it is common to plot solutions corresponding to points that are very close in the plot and connect the points via a solid straight or curved line. As it will be seen later for functions, there are ways to estimate what is the most appropriate curvature to sketch, also called **plot**, the graph of the function.

2) Origins of a equation: practical examples

There are different ways an statement can originate other than the “funs” of putting together a puzzle. The first and perhaps most exciting one is to find a quantitative property that your target set of values, and only them, have. Such a quantitative property will translate into a mathematical statement and the target values will be its solutions. Naturally, for this to work, it is necessary to be possible to translate such a quantitative property into a mathematical statement within the scope of the mathematical operations defined. Some examples for equations containing 2 or more variables are shown below. Try doing them !

Exercise 1: Two dices will be thrown. I would like to reference all the possibilities of resulting values that, put together, would yield a value of seven. But I don't want to list all the possibilities by hand. Instead, I want to write an equation that I am sure will implicitly specify it for me. Are you able to formulate such an equation?

Exercise 2: There is a circumference, which, put in a Cartesian coordinates system, is centred at $(3, -2)$ and has a radius equal to three. I would love to write an equation to reference all the points that are part of the circumference in the scope of the real numbers. Are you able to think of an equation that would be solved by all of these points and only them? Tip: A good way would be to ask yourself beforehand if you are able to spot a property the points making a circumference have in common and if this property can be translated into a mathematical statement.



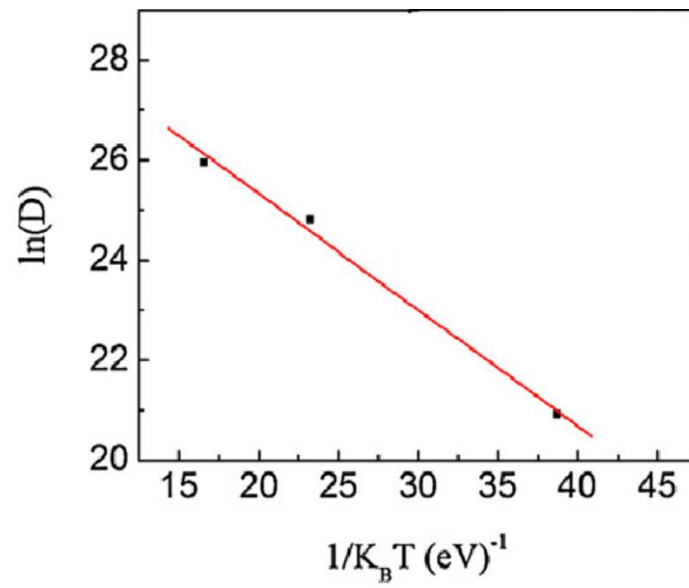
The second way of deriving an equation that is often used in science is from having a sample of combinations of values of variables that are known to be true and then try to find the algebraic pattern that best reproduce this occurring combination of values observed. This is useful particularly when there is no prior knowledge of a quantitative property shared by the occurring combinations of values. Generally, the idea is to attempt

to **fit** equations of different forms (polynomials, trigonometric, etc), with each form having specific parameters to-be-determined in the fitting and also involving the variables concerned, into the data you have and quantitatively evaluating which form allows to better correlate the variables.² The fitting finds the best parameters within the given function form that reproduces with minimum error the occurring combination of variable values that you know. For example, for two variables we are calling x and z , we may try to fit a trigonometric function of the form $x = a\cos(z) + b$, where a and b are parameters (constants), in a set of combinations of values of x and z that are known to occur. The fitting procedure would yield the values of a and b that most optimally reproduce the occurring combinations as well as a parameter, symbolized here as R^2 , that quantifies how far the solutions predicted by the relationship are far from the occurring ones. Within the previous example, this could be determined from the absolute difference between the value of x obtained for a given z and the value of x you have for that given z in the dataset.

The fundamentals of a fitting procedure can be learnt elsewhere and it is implemented in many computer programs. The idea is that fitting many equation forms can eventually allow you to reach one that satisfyingly reproduce the occurring combinations. It is also interesting to note however that in this scenario there is no guarantee that the statement you found in the fit holds true for solutions other than the ones you know: one can only hope. Assuming it holds for other points within the interval comprised within the previously known combination of variable values was made is called **interpolating**. Assuming the relation holds in intervals outside the previously known combination of variable values is called **extrapolating**. The example below shows an example of this scenario where the variables are actually physical properties that are correlated. It is worth saying that this is often the case that this is the way we discover quantitative relationships within science.

Example 3: The self diffusion coefficient (D), a property linked to the long-term movement of a given chemical species, of 13-atom Cu cluster on silver was estimated experimentally at different temperatures (T). Several correspondence between occurring values of $\ln(D)$ and $1/(k_B T)$, where k_B is a constant, were collected. Three occurring values are shown graphically in the figure below (instead of points, squares are used). It is worth noting that as in any experimental result, there is always an error associated to measurement, but we do not care about this here and rather regard the solutions as the rightful occurring combinations of $\ln(D)$ and $1/(k_B T)$. An attempt was made to find the best equation that relates two variables, x and y where $x \equiv \ln(D)$ and $y = 1/(k_B T)$. The resulting equation is $y = 29.97 - 232.06x$ and has a graphical representation indicated by the straight line shown below in the range of $x \in [15, 41]$. It is possible to visualize that the errors are not expected to be huge and that the equation found reasonably reproduces the experimental data.

² To think: have we defined enough operations to be able to, by using them, reproduce whichever pattern exists?



Source : Sardar Sikandar Hayat. Pop-up of atoms among copper 13-atom island on Ag(1 1 1). Computational Materials Science, v. 50, p. 1485-1489 (2011).